MATH 152 **QUIZ 5** NAME _____

Mrs. Bonny Tighe 25 points

7.7 TA_____ Wed 10/9/2019

1. Find the error bounds for all three methods. $\int_1^2 \sqrt{x}\,dx, \quad n = 4$

Suppose $|f''(x)| \le K \quad for \quad a \le x \le b$, then $|E_T| \le \dfrac{K(b-a)^3}{12n^2}$ $\quad |E_M| \le \dfrac{K(b-a)^3}{24n^2}$

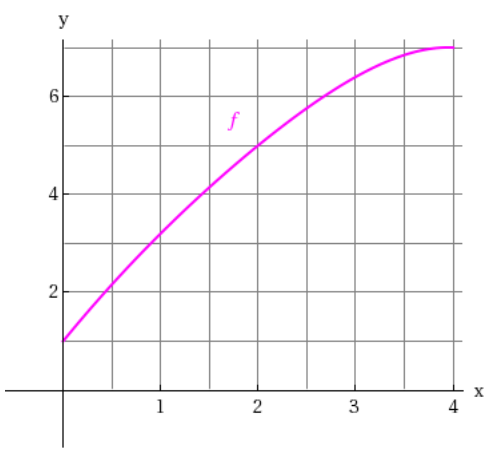
Suppose $|f''''(x)| \le K \quad for \quad a \le x \le b$ then $|E_S| \le \dfrac{K(b-a)^5}{180n^4}$

2. A radar gun was used to record the speed of a runner during the first 5 seconds of a race (see the table). Use Simpson's Rule to estimate the distance the runner covered during those 5 seconds.

t (s)	v (m/s)		t (s)	v (m/s)
0	0		3.0	12
0.5	4		3.5	10
1.0	7		4.0	11
1.5	8		4.5	10
2.0	9		5.0	9
2.5	11			

3. Use (a) the Trapezoidal Rule and (b) Simpson's Rules to approximate the given integral with the specified value of n for $\int_0^{\pi/2} \sin 2\theta \, d\theta, \quad n = 4$

4. $I = \int_0^4 f(x)dx$ where f is the function whose graph is shown.



(a) Use the graph to approximate $M_4, \; L_4, \; R_4$

5. How large should we take n in order to guarantee that Simpsons approximation for $\int_1^2 \frac{1}{x^2} dx$ is accurate to within 0.001?

MATH 152 **QUIZ 5A** NAME _____

Mrs. Bonny Tighe 25 points

7.7 TA_____ Wed 10/9/2010

1. Find the error bounds for all three methods. $\int_2^3 \frac{1}{x}\,dx, \quad n=4$

Suppose $|f''(x)| \le K \text{ for } a \le x \le b$, then $|E_T| \le \frac{K(b-a)^3}{12n^2}$ $|E_M| \le \frac{K(b-a)^3}{24n^2}$

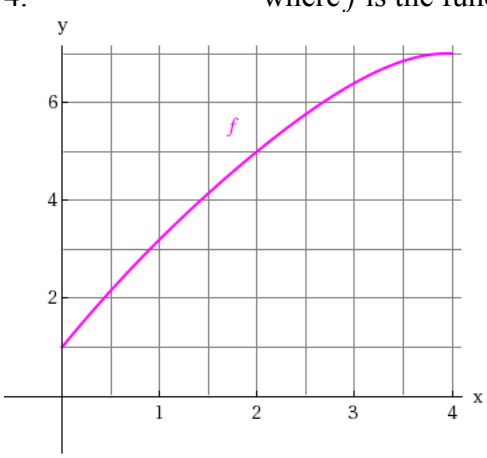
Suppose $|f''''(x)| \le K \text{ for } a \le x \le b$ then $|E_S| \le \frac{K(b-a)^5}{180n^4}$

2. The table shows values of a force function $f(x)$, where x is measured in meters and $f(x)$ in newtons. Use Simpson's Rule and the Trapezoid Rules to estimate the work done by the force in moving an object a distance of 18 m.

x	0	3	6	9	12	15	18
$f(x)$	9	9	8	8	7	7	7

3. Use (a) the Trapezoidal Rule and (b) Simpson's Rules to approximate the given integral with the specified value of n for $\int_0^{\pi/2} \cos 2\theta \, d\theta, \quad n=4$ Evaluate any trig functions.

4. $I=\int_0^4 f(x)\,dx$ where *f* is the function whose graph is shown.



(a) Use the graph to approximate $M_4,\ L_4,\ R_4$

5. How large should we take n in order to guarantee that Simpsons approximation for $\int_1^2 \sqrt{x}\,dx$ is accurate to within 0.001?